

teristics. Fibrin hydrogel materials and alginate-like polymers are mostly being used in printing. Other naturally-derived polymers include gelatin, collagen, fibrin, chitosan and hyaluronic acid.

Some synthetic polymers include polyethylene glycol, polycaprolactone, polylactic acid, polyglycolic acid and poly (lactic-co-glycolic) acid. A 3D extracellular matrix helps in preserving the native tissue architecture.

The machine that prints humans. The typical system, known as integrated tissue and organ printing system (ITOP), was developed into the current 3D bio-printer we have today over a period of ten years. The technology used in the development of bio-printers is another interesting one. The idea to heal the most difficult of wounds came from inkjet printers.

Experts from Wake Forest Institute explain the schematics of the machine. Multiple cartridges are used to deliver and pattern multiple cell-laden composite hydrogels, supporting thermoplastic polymer and a sacrificial removable hydrogel. End of each cartridge is connected to a micro-scale nozzle, and the top is connected to an air-pressure controller for precisely controlling the dispensing volume.

A heating unit ensures that the melted polymer remains easily dispensable. A three-axis motorised stage system enables 3D patterning of multiple cells and bio-absorbable materials.

The system resides in a humidified and temperature-controlled enclosure

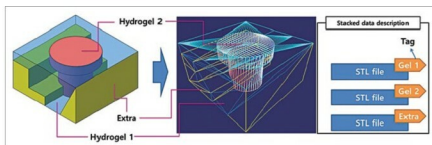


Fig. 2: Bio-printing, CAD process at POSTECH

with clean environment. It works by depositing the hydrogel that contains cells, along with the scaffold material.

The challenge here is to ensure the life of the tissue structure is long enough for successful integration with the human body. This issue is addressed by a lattice structure of micro-channels that allow nutrients and oxygen from the body to pass into the graft until the tissue regenerates its own blood vessels.

Growth in systems

Dispensing based 3D printing has been suggested for direct 3D cell printing of cell-laden hydrogel constructs with a larger volume. Although this system generally has low resolution for making a cell pattern, dispensing of hydrogel solution with a high flow rate allows more rapid printing of 3D cell-laden hydrogel constructs. Major developments in the ITOP system have been related to nozzles and dispensing modules that can print at a higher resolution.

Researchers at Pohang University of Science and Technology (POSTECH) in South Korea have reported a computer aided design

(CAD) and manufacturing system for multiple-head 3D printing, and have printed heterogeneous tissue models using two kinds of cell-laden hydrogel.

CATIA V5, a 3D CAD design software from Dassault Systemes, is being used by researchers at POSTECH in their design process. A paper published by Jin Woo Jung and Jung-Seob Lee, researchers at Department of Mechanical Engineering, POSTECH, explained the use of CATIA V5 for designing the exterior structure of heterogeneous tissues. The resulting models had two main tissue parts, plus a support part.

3D free-form structures are printed via vertical stacking of 2D patterns generated from cross-sections of the designed 3D model. The slicing algorithm involves calculating contours of the cross-sections of successive planes. Printing path for the frame is defined based on cross-section contours of the two cell-laden hydrogel parts.

Scaffold-free printing is here. Scaffold-free printing is another area that is catching up in recent years. Dr Gabor Forgacs, founder of Organovo and Modern Meadow, with his academic partners, leveraged the understanding of mammalian cells, to come up with another interesting system. They were able to demonstrate directed assembly of 3D tissues without dependency on integrated scaffolding or biomaterial components.

Similar results have been published in a paper on scaffold-free tubular tissues created by a bio-3D

The printing process

Once you have the required dimensions in order, a virtual frame for the organ is generated in a CAD program. The CAD design is then loaded to the bio-printer head pin, filled with hydrogel to generate the organ structure. Then, stem cells come into the picture, depending on the organ.

Printing is done layer by layer. First, a layer of hydrogel is printed as a frame, followed by differentiated stem cells. Printing is done from the inside out, to leave cavities of a thickness ten times lower than human hair.

Blood runs through these cavities bringing organs to life. The process of printing an organ can be segmented and can take up to ten hours for a right knee cartilage and up to 48 hours for an oesophagus.

Printed parts are then kept in a bioreactor where these are fed and cultivated to help their cells replicate in dozens of millions.